



YEAR 11
MATHEMATICS EXTENSION 1 ASSESSMENT
JUNE 2008
TIME : 70 minutes

Directions:

Full working should be shown in every question.

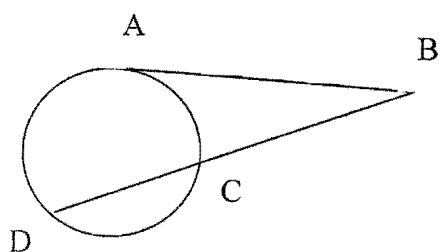
Marks may be deducted for careless or badly arranged work.

Use black or blue pen only (not pencils) to write your solutions.

No liquid paper is to be used.

If a correction is made, one line is to be ruled through the incorrect answer.

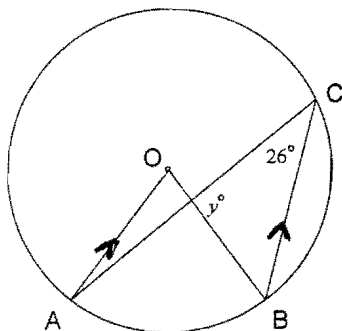
- | | Marks |
|---|--------------|
| 1. Solve $\frac{2x+3}{x-2} \leq 1$ | 3 |
| 2. Find, as an exact value $\sin 75^\circ$, showing working. | 3 |
| 3. Find the coordinates of the point which divides the interval joining A(3, -4) And B(-2, 3) externally in the ratio 2 : 3 | 2 |
| 4. a) AB is a tangent to the circle.
DC is 5cm and CB is 4cm. Find the length of AB. | 1 |



NOT TO SCALE

- b) Find the size of y

1



NOT TO SCALE

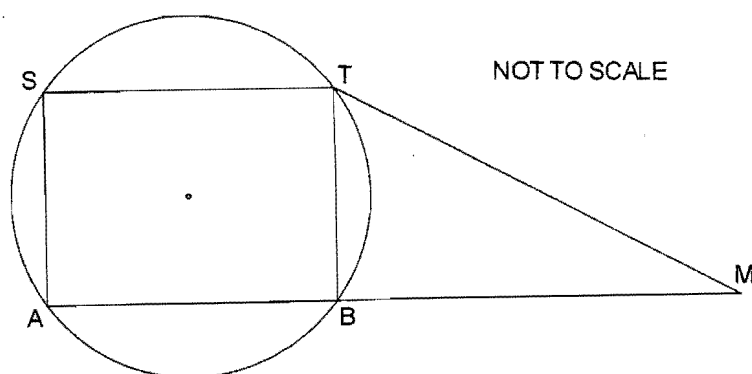
5. Find the acute angle between the lines $x - 2y = 0$ and $3x - y - 15 = 0$

3

6. Solve $\sin^2 \theta + 2 \sin \theta \cos \theta - 3 \cos^2 \theta = 0$ for $0^\circ \leq \theta \leq 360^\circ$

4

7.



Given $ST \parallel AB$ and TM is a tangent to the circle

i) Prove $\triangle TMB \sim \triangle TAS$

3

ii) If $BM = 4$, $SA = 5$ and $TB = 7$, find the length of ST

2

8. α and β are acute angles such that $\cos \alpha = \frac{3}{5}$ and $\sin \beta = \frac{1}{\sqrt{5}}$

3

Without finding the size of either angle, show that $\alpha = 2\beta$

9. Prove the trig identity

3

$$\frac{\sin x}{\sin 2x} + \frac{\cos x}{1 + \cos 2x} = \sec x$$

10. Express $\cos x - \sqrt{3} \sin x$ in the form

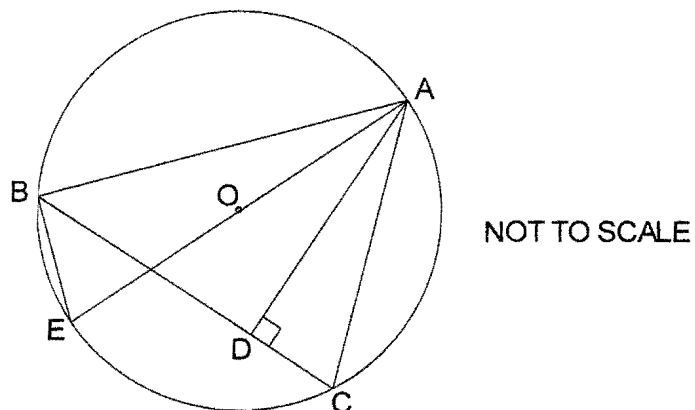
i) $A \cos(x + \alpha)$ where $A > 0$, $0 < \alpha < 90^\circ$

2

ii) Hence solve $\cos x - \sqrt{3} \sin x = 1$ for $0^\circ \leq x \leq 360^\circ$

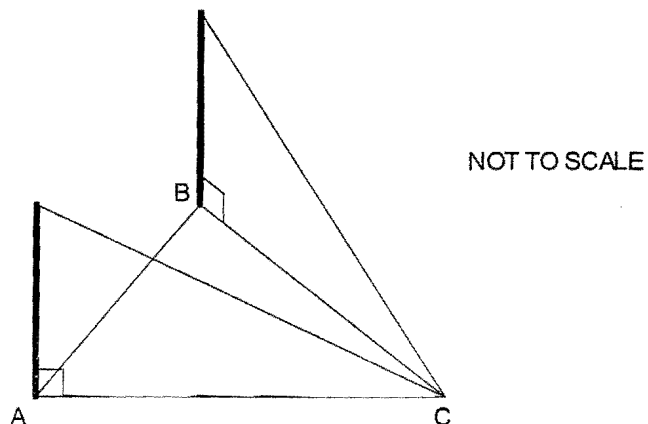
2

11. ABC is a triangle inscribed in a circle, centre O, and AD is drawn perpendicular to BC. AE is the diameter. 4



Prove $\angle BAE = \angle DAC$

12. A and B are the feet of two towers of equal heights. B lies due North of A. From a point C, 40m East of A and in the same horizontal plane, the angle of elevation at the top of the tower A is 53° . From the same point, the angle of elevation of tower B is 35° .

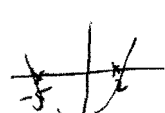


- i) Show that $AB = \frac{40\sqrt{\tan^2 53 - \tan^2 35}}{\tan 35}$ 3
- ii) Hence find the distance AB to the nearest metre. 1
13. Prove the identity $\frac{\tan 2\theta - \tan \theta}{\tan 2\theta + \cot \theta} = \tan^2 \theta$ 3

Solutions yr 11 Ext 1. Exam June 2008.

1. $\frac{2x+3-x+2}{x-2} \leq 0$

$$(x-2)^2 \frac{(x+5)}{(x-2)} \leq 0 \times (x-2)^2 \quad \text{① for working}$$

$$\therefore (x-2)(x+5) \leq 0$$


$$-5 \leq x < 2$$

but $x \neq 2$ ①

$$\therefore \boxed{-5 \leq x < 2} \quad \text{①} \quad \boxed{3}$$

2. $\sin 75 = \sin(45+30)$ ①

$$= \sin 45 \cos 30 + \cos 45 \sin 30$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2} \quad \text{①}$$

$$= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}$$

$$= \text{①} \frac{\sqrt{3}+1}{2\sqrt{2}} \text{ or } \frac{\sqrt{6}+\sqrt{2}}{4} \quad \boxed{3}$$

3. $\begin{matrix} x_1 & y_1 & x_2 & y_2 & m & n \\ (3, -4) & (-2, 3) & 2 & 3 \end{matrix}$

$$x = \frac{m x_2 - n x_1}{m - n} \quad y = \frac{m y_2 - n y_1}{m - n}$$

$$= \frac{2 \times -2 - 3 \times 3}{2 - 3} \quad = \frac{2 \times 3 - 3 \times -4}{2 - 3}$$

$$= \frac{-13}{-1} \quad = \frac{18}{-1}$$

$$\therefore \text{①} \quad \text{①} \quad (13, -18) \quad \boxed{2}$$

4. a) $AB^2 = DB \times CB$

$$= 9 \times 4$$

$$= 36$$

$$AB = 6 \text{ cm} \quad \text{①}$$

b) $y = 180 - (26 + 52)$

$$= \underline{102^\circ} \quad \text{①} \quad \boxed{2}$$

5) $l_1: y = \frac{1}{2}x \quad l_2: y = 3x - 15$

$$m_1 = \frac{1}{2} \quad \text{①} \quad m_2 = 3$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{1}{2} - 3}{1 + \frac{1}{2} \times 3} \right| \quad \text{①}$$

$$= 1$$

$$\theta = 45^\circ \quad \text{①} \quad \boxed{3}$$

6) $(\sin \theta + 3 \cos \theta)(\sin \theta - \cos \theta) = 0$ ①

$$\sin \theta = -3 \cos \theta \quad \text{①} \quad \sin \theta = \cos \theta$$

$$\tan \theta = -3 \quad \tan \theta = 1 \quad \text{①}$$

$$\theta = 108^\circ 26', 288^\circ 26' \quad \theta = 45^\circ, 225^\circ$$

$$\therefore \theta = 45^\circ, 108^\circ 26', 225^\circ, 288^\circ 26' \quad \boxed{4}$$

7) In $\triangle TMB$ and $\triangle TAS$

i) $\hat{A}ST = \hat{T}BM$ (ext L of cyclic quad equal to inter. opp L) ①

$\hat{M}TB = \hat{T}AB$ (L between tangent and chord at pt of contact equal to L in alternate segment) ①

since $ST \parallel AB$ (given)

then $\hat{S}TA = \hat{T}AB$ (alternate Ls, equal)

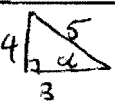
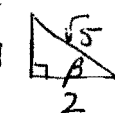
$$= \hat{M}TB$$

$\therefore \triangle TMB \parallel \triangle TAS$ (matching angles equal) ①

ii) $\frac{ST}{TB} = \frac{SA}{BM}$ (matching sides in ratio)

$$\therefore \frac{ST}{7} = \frac{5}{4} \quad \text{①}$$

$$ST = \frac{35}{4} \quad \text{①} \quad \boxed{5}$$

8. $\cos \alpha = \frac{3}{5}$  $\sin \beta = \frac{1}{\sqrt{5}}$  ①

$$\sin \alpha = \frac{4}{5}$$

$$\sin 2\beta = 2 \sin \beta \cos \beta$$

$$= 2 \times \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}} \quad \text{①}$$

$$= \frac{4}{5} \quad \text{①}$$

$\therefore \sin \alpha = \sin 2\beta$ then $\alpha = 2\beta$

$$9. \frac{\sin x}{\sin 2x} + \frac{\cos x}{1 + \cos 2x} = \sec x$$

$$\text{LHS} = \frac{\sin x}{2 \sin x \cos x} + \frac{\cos x}{1 + 2 \cos^2 x - 1}$$

$$= \frac{1}{2 \cos x} + \frac{\cos x}{2 \cos^2 x}$$

$$= \frac{1}{2 \cos x} + \frac{1}{2 \cos x}$$

$$= \frac{2}{2 \cos x} = \frac{1}{\cos x}$$

must have
= RHS

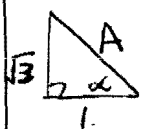
$$= \sec x = \text{RHS.} \quad [3]$$

$$10. \cos x - \sqrt{3} \sin x.$$

$$i) A \cos x \cos \alpha - A \sin x \sin \alpha.$$

$$1 = A \cos \alpha \quad \sqrt{3} = A \sin \alpha$$

$$\cos \alpha = \frac{1}{A} \quad \sin \alpha = \frac{\sqrt{3}}{A}$$

 $A = 2 \quad \tan \alpha = \sqrt{3}$
① $\alpha = 60^\circ$ ①

$$\therefore \cos x - \sqrt{3} \sin x = \frac{2 \cos(x + 60^\circ)}{2}$$

$$ii) 2 \cos(x + 60) = 1 \quad 60 \leq x + 60 \leq 420$$

$$\cos(x + 60) = \frac{1}{2} \quad ①$$

$$x + 60 = 60, 300, 420$$

$$x = 0, 240, 360 \quad ①$$

[4]

$$ii) \text{Let } \hat{BAE} = x \quad ①$$

$$\hat{ABE} = 90^\circ \text{ (L in a semi circle)} \quad ①$$

$$\therefore \hat{BEA} = 90 - x \text{ (L sum of } \triangle ABE) \quad ①$$

$$\hat{BCA} = 90 - x \text{ (Ls at circum. standing on same arc are equal)} \quad ①$$

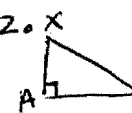
$$\hat{ABC} = 90 \text{ (given)}$$

$$\therefore \hat{DAC} = 180 - (90 - (90 - x)) \text{ (L sum } \triangle DAC) \quad ①$$

$$= x$$

$$\therefore \hat{BAE} = \hat{DAC}$$

[4]

12.  In $\triangle ACX$
 $\tan 53 = \frac{h}{40}$
① $h = 40 \tan 53$

$\therefore$ In $\triangle ABC$ $\hat{BAC} = 90^\circ$

$$\therefore BC^2 = AB^2 + AC^2$$

$$AB^2 = BC^2 - AC^2$$

$$= \left(\frac{40 \tan 53}{\tan 35} \right)^2 - (40)^2$$

$$= \frac{40^2 \tan^2 53 - 40^2 \tan^2 35}{\tan^2 35} \quad ①$$

$$AB = \frac{40 \sqrt{\tan^2 53 - \tan^2 35}}{\tan 35}$$

$$ii) AB = 64.396 \dots$$

$$= \underline{\underline{64 \text{ m}}} \quad ①$$

[4]

$$13. \frac{\tan 2\theta - \tan \theta}{\tan 2\theta + \cot \theta} = \tan^2 \theta$$

$$\text{LHS} = \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} - \tan \theta \right) \div \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} + \frac{1}{\tan \theta} \right)$$

$$= \frac{2 \tan \theta - \tan \theta + \tan^3 \theta}{(1 - \tan^2 \theta)} \times \frac{(1 - \tan^2 \theta)}{(2 \tan \theta + 1 - \tan^2 \theta)}$$

$$= \frac{\tan \theta (1 + \tan^2 \theta) \times \tan \theta}{(\tan^2 \theta + 1)} \quad ①$$

$$= \tan^2 \theta$$

$$= \text{RHS} \quad \text{no mark here refer (9)} \quad \text{(maybe easier ① off each error)}$$

[3]